Machine Learning Welcome

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Course Syllabus

- Learning Types
  - Supervised:
    - Regression
    - Classification
- K-nearest Neighbors
- Decision Tree
- Bayesian Classifier
- Logistic Regression
- Support Vector Machines
- Ensemble Methods

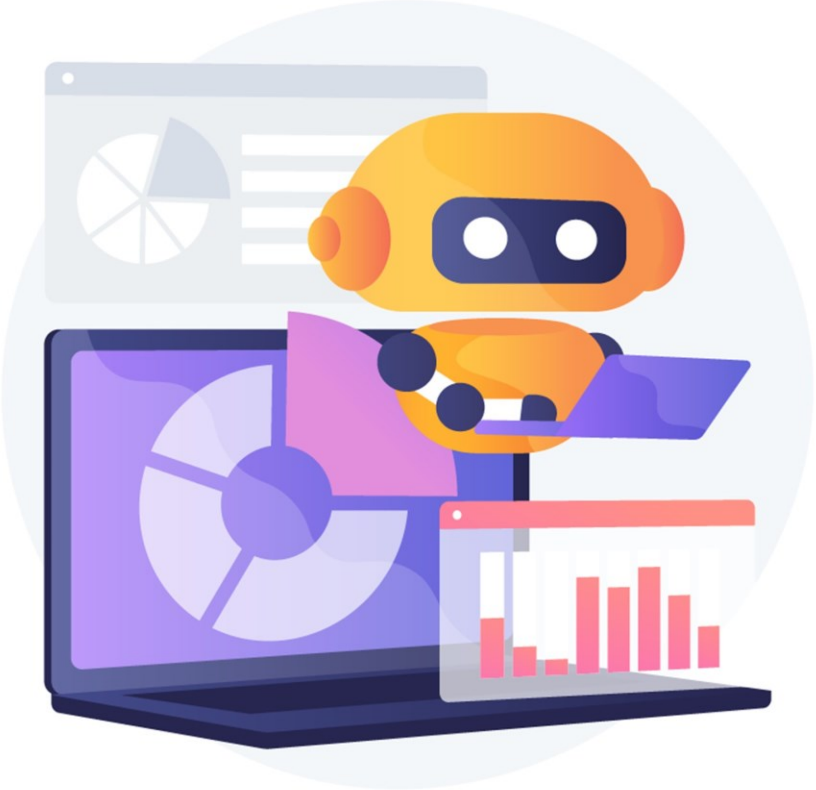
References

- Yaser S. Abu-Mostafa et al., Learning from data, AMlBook, 2012.
- Ethem Alpaydin , An Introduction to Machine learning, MIT Press, 3rd ed., 2014.
- Kevin Murphy, Machine Learning: a Probabilistic Perspective, MIT Press, 2012.

- Stuart Russell, Peter Norvig, Artificial Intelligence: A Modern Approach, 4th edition, Pearson, 2021.
- Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
- Tom Mitchell, Machine Learning, McGraw-Hill, 1997.
- Some recent or seminal papers.

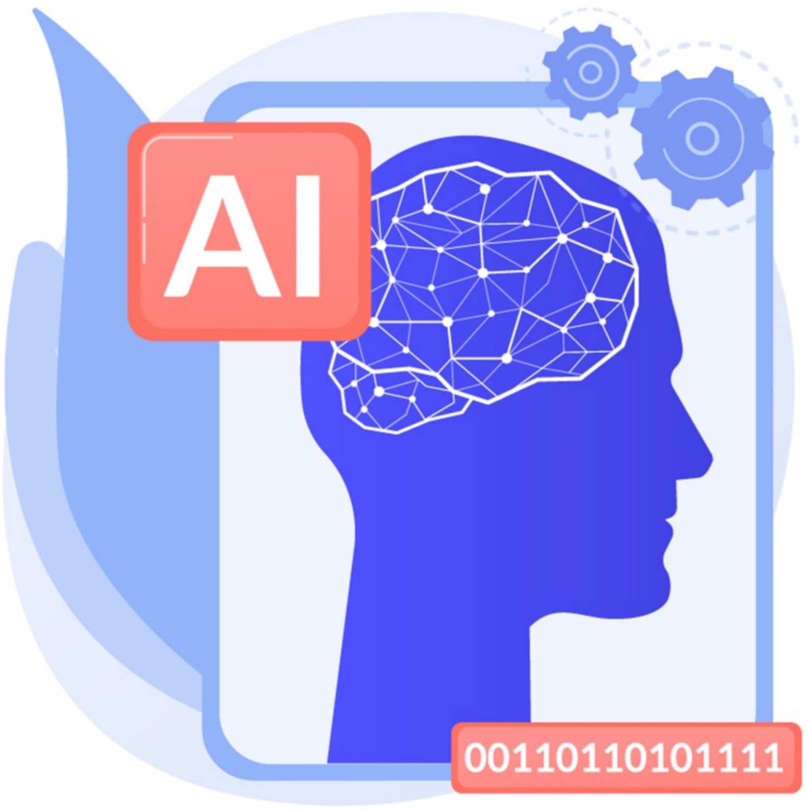
چهار شاخه مرتبط

- Statistics
- Artificial Intelligence
- Machine Learning
- Data Mining



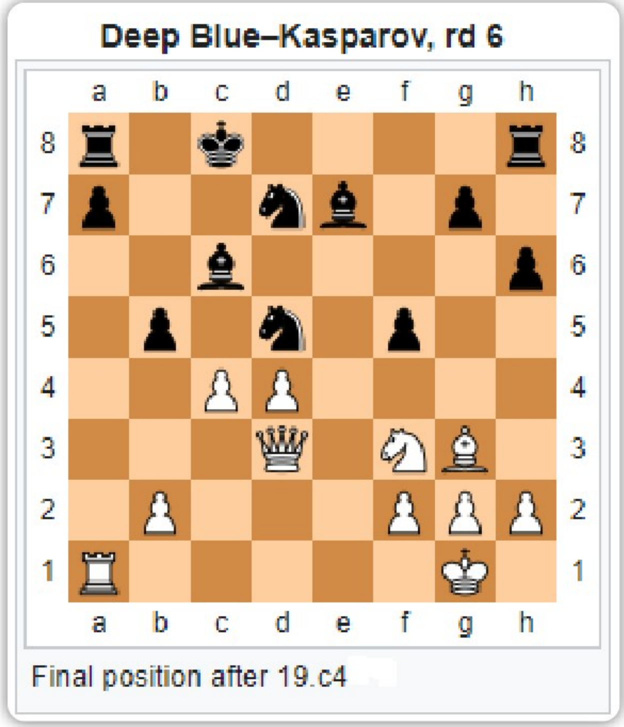
Artificial Intelligence

- Main AI problems:
  - Reasoning
  - Knowledge representation
  - Automated planning
  - Machine learning
  - Natural language processing
  - Computer vision
  - Robotics
  - Chatbots:
    - ALICE
    - Oliver



When AI Beats Humans

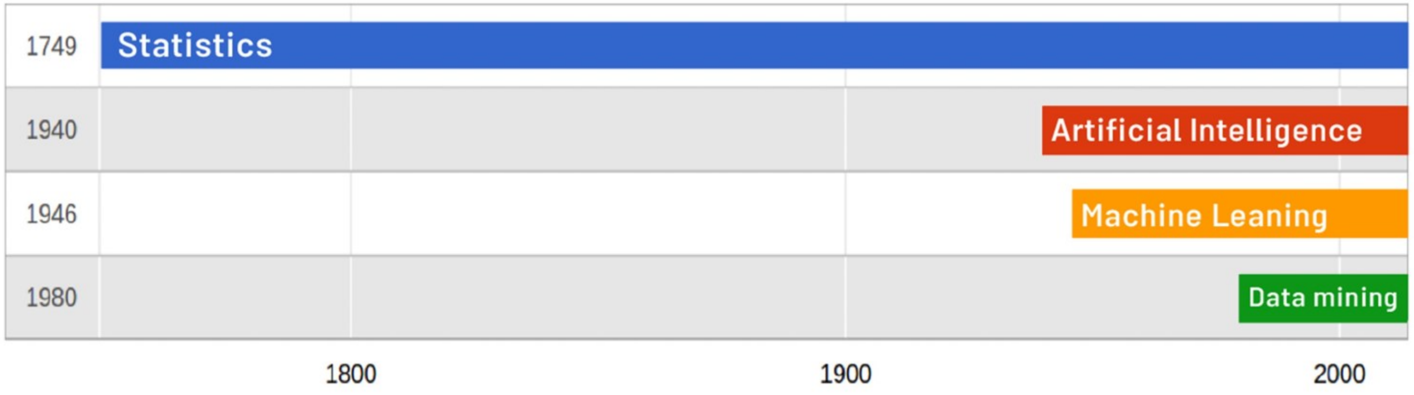
Deep Blue, 1997



AlphaGo, 2016

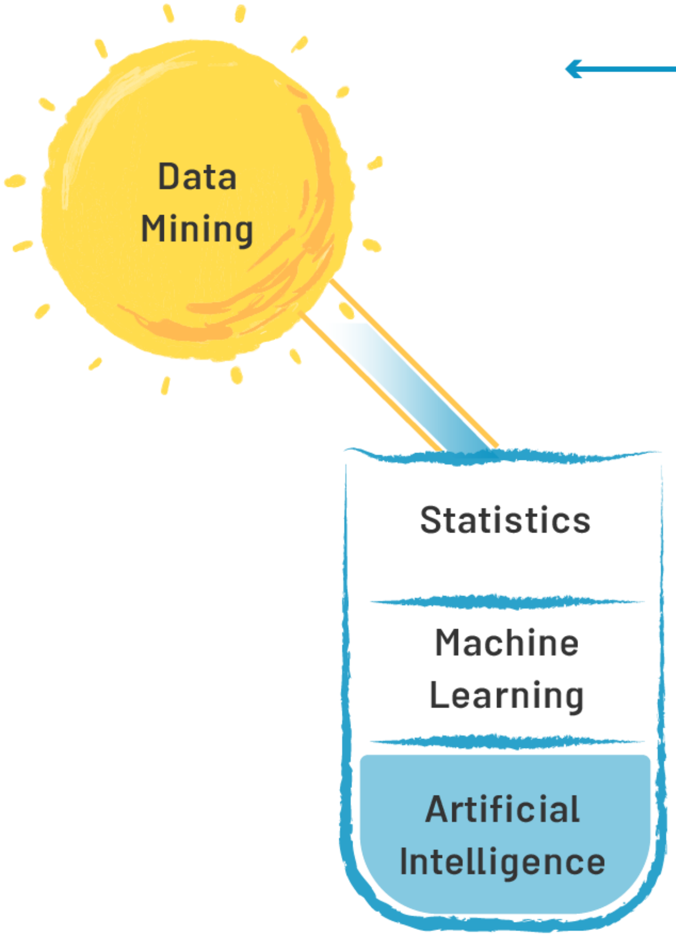
Hierarchy

سلسه مراتب



Interdependency

ارتباط چهار شاخه



What is Machine Learning?

- Herbert Simon (1970)
  - Any process by which a system improves its performance.
- Tom Mitchell (1990)
  - A computer program that improves its performance at some task through experience.
- Ethem Alpaydin (2010)
  - Programming of computers to optimize a performance criterion using example data or past experience.
- Shai Ben-David (2014)
  - Automated detection of meaningful patterns in (large) datasets.

Machine Learning (cont.)

- Machine learning (ML) develops algorithms for making predictions from data.
  - What does "prediction" mean here?

When ML is Used?

- Three criteria:
  - 1. A pattern exists.
  - 2. We cannot pin it down mathematically.
  - 3. We have data on it ++

Learning Types

- Types of machine learning algorithms:
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning

    - - -

    - Semi-supervised learning
    - Self-supervised learning
    - Deep Learning
    - Transfer learning
    - Active learning
    - Multilabel prediction
    - ...

Machine Learning Scholars

Robert Tibshirani — Cited by 277501

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Statistics Applied Statistics Statistical learning machine learning data science

Geoffrey Hinton — Cited by 239729

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machine learning neural networks artificial intelligence cognitive science computer science

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machine learning statistics computer science

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machine learning · statistics · computational biology · artificial intelligence · optimization

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Machine learning · deep learning · artificial intelligence

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AI · Image Processing · Computer Vision · Pattern Recognition · Machine Learning

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Director, Max Planck Institute for Intelligent Systems; and Distinguished Amazon Scholar
Venfied email at tuebingen.mpg.de

Machine Learning · Causal Inference · Artificial Intelligence · Computational Photography · Statistics

Applications

- Image & Speech Recognition
- Natural Language Processing
- Handwriting Recognition
- Autonomous Control (Planes, Trains, Automobiles, ...)
- Anomaly Detection in E-commerce (Fraud Detection)
- Twitter Topic Modeling
- Targeted Marketing
- Medical Diagnosis
- Rank Webpages in Search Engines
- Email Spam Filtering
- Recommender Systems
- Bioinformatics
- Sleep Quality Detection

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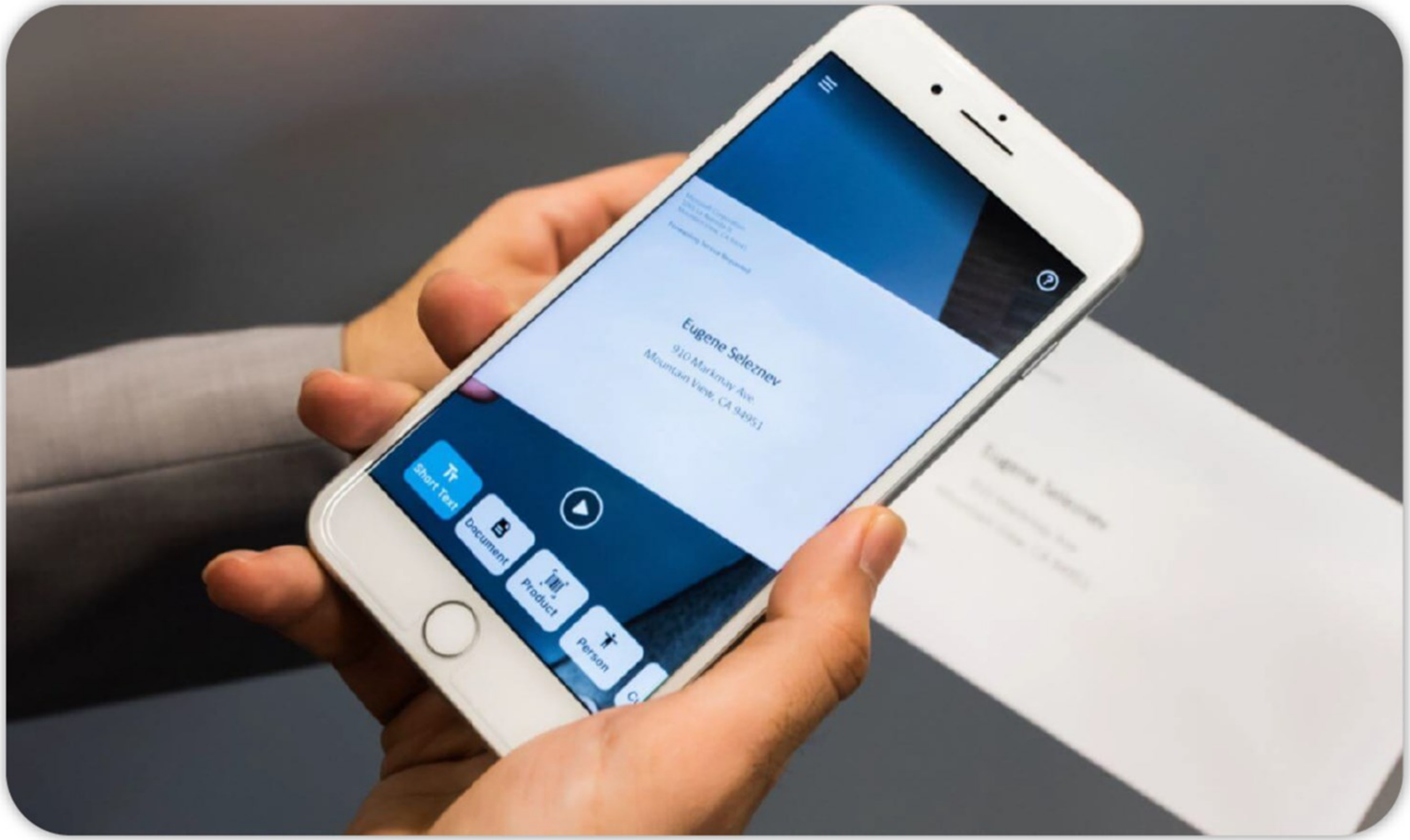
Netflix Prize

IBM Watson

Microsoft Seeing AI Project



Robotics

Which of the following choices matches this figure?

- A. ROBI
- B. NAO
- C. ASIMO
- D. PEPPER
- E. CHEETAH

References

- Question & answer websites, including (not limited to):
  - shakthydoss.com
  - www.stackoverflow.com
  - stats.stackexchange.com
  - www.quora.com
  - ...